

BE/Insem/APR-26

B.E. (Mechanical) (Semester - II)

FINITE ELEMENT ANALYSIS

(2012 Pattern) (Elective - IV)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4 and Q.5 or Q.6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.

- Q1)** a) Discuss the advantages and disadvantages of finite element method over conventional methods. [4]
- b) Explain the terms 'plane stress' and 'plane strain' problems used in FEA using suitable examples. [6]

OR

- Q2)** a) What is mesh refinement? Explain h-refinement and p-refinement. [5]
- b) Explain Rayleigh Ritz approach used in FEA. [5]

- Q3)** a) Derive the expression for shape function for a two noded bar element taking natural coordinate & as varying from -1 to 1. [4]
- b) A system of spring in shown in Fig. 1 find [6]
- i) Nodal displacement.
 - ii) Reaction forces.

$$K_1 = 10 \text{ N/mm} \quad K_2 = 15 \text{ N/mm} \quad K_3 = 25 \text{ N/mm} \quad K_4 = 20 \text{ N/mm}$$

$$F_2 = 20 \text{ N}, \quad F_3 = 30 \text{ N}, \quad F_4 = 50 \text{ N},$$

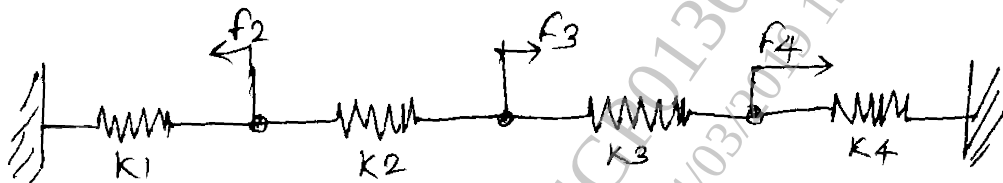


Fig. 1.

OR

P.T.O.

- Q4) a)** Using direct stiffness method, determine the nodal displacement of stepped bar shown in Fig. 2. [6]

Take $E_1 = 200 \text{ GPa}$ $E_2 = 70 \text{ GPa}$
 $F_1 = 10 \text{ kN}$ $F_2 = 5 \text{ kN}$
 $A_1 = 150 \text{ mm}^2$ $A_2 = 100 \text{ mm}^2$

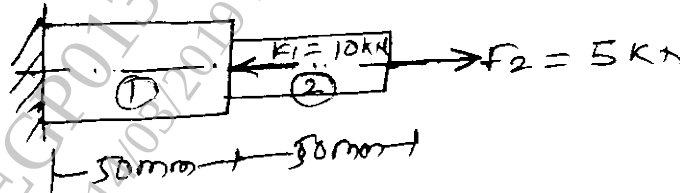


Fig. 2.

- b) Derive formulation of Element Stiffness Matrix and load vector for Truss. [4]

- Q5) a)** The triangular element has nodal coordinates (10, 10) (40, 20) and (30, 50) for nodes 1, 2 and 3 respectively for a point 'p' inside triangle. Determine the x and y coordinates if shape functions N_1 and N_2 are 0.15 and 0.25 respectively. [6]

- b) Write short note on CST element and LST element. [4]

OR

- Q6) a)** Write short note on axi-symmetric elements. [4]

- b) Explain properties of shape functions of 2 D elements. [6]

